Experiment 4: the moment of inertia of an iOLab, from energy and from torque

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Purpose

Engineers who test scale model windmills fix an iOLab to the wings, so the iOLab adds its own moment of inertia to the rotating wings. To correct their measurements they need the moment of inertia of the iOLab about its y axis. We determine it in two parts: Part 1 from conservation of energy and Part 2 from Newton's second law. In both, the iOLab with a tape roll on each end rolls down a ramp while its gyroscope measures the angular velocity.

1 Setup and measurements

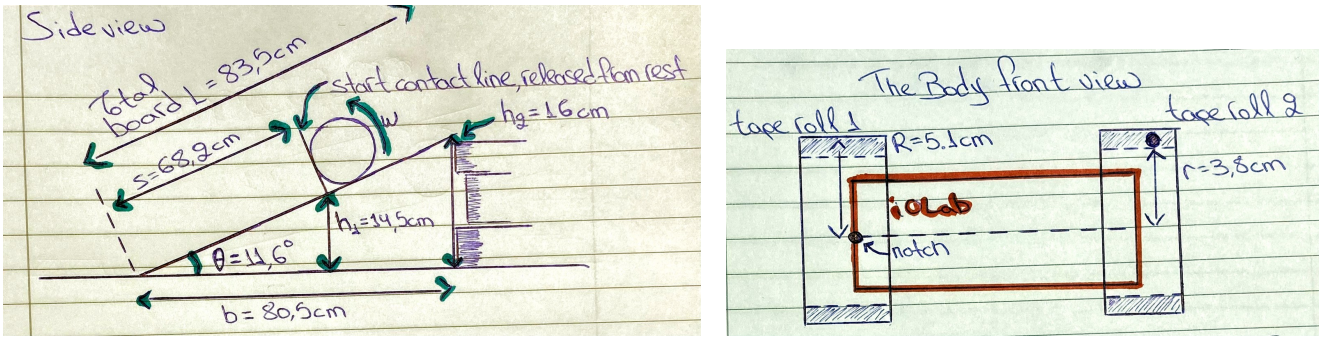


Figure 1. Our hand drawn sketch of the setup. Left: the ramp from the side, angle drawn larger than real; h_1 is the height of the board at the start line, h_2 at the books. Right: the body from the front. The tape rolls sit on the ends of the iOLab up to the notches, so the body turns about the y axis of the iOLab.

A tape roll was pressed onto each end of the iOLab up to the notches (Figure 1). The body rolls on the outer surface of the two rolls and turns about the y axis of the iOLab, so the gyroscope measures the angular velocity directly as ω_y. The board, $L = 83.5$ cm long, rested with its low end on the table and its high end on books. Both rolls are the same kind of packing tape and differ only very slightly. Table 1 lists the measured inputs; ω and α count positive when the body rolls down.

Table 1. Measured inputs. Uncertainties: the last digit of the scale, and the reading of the folding rule and the tape measure.

Quantity	Value	How	Quantity	Value	How
m_{iOLab}	(201.89 ± 0.01) g	lab scale	θ	$(11.6 \pm 0.2)^\circ$	phone app on the board
m_{rolls}	(352.62 ± 0.01) g	both rolls	$\sin\theta$	0.2011 ± 0.0034	
M	(554.51 ± 0.02) g	sum	s	(68.2 ± 0.2) cm	contact line to the end
R	(5.10 ± 0.05) cm	outer diameter 10.2 cm	L, b	83.5, 80.5 ($\pm$0.3) cm	board, along the table
r	(3.80 ± 0.05) cm	hole diameter 7.6 cm	h_1, h_2	14.5, 16.0 ($\pm$0.2) cm	start line, books
I_{rolls}	$(7.1 \pm 0.3) \times 10^{-4}$ kg m^2	eq. (4)	f	190 Hz	gyroscope rate

A phone app laid on the board gave $\theta = 11.6^\circ$. The lengths in Figure 1 give 11.1° ($\sin\theta = h_2/L$), 11.2° ($\tan\theta = h_2/b$) and 12.3° ($\sin\theta = h_1/s$). They scatter around the phone value: a height that is 1 cm off moves them by about 0.8°, so we use the phone value.

2 Theory

The body rolls without slipping, so $v = \omega R$ and $a = \alpha R$. The static friction force f from the board acts at the contact line, which is at rest, so f does no work and drops out when we eliminate it in (3). I is the moment of inertia of the whole body about its own axis, the y axis of the iOLab.

Part 1, conservation of energy. From rest the body rolls the distance s along the board and drops $h = s\sin\theta$. The potential energy becomes kinetic energy of translation and rotation:

$$Mgh = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}I\omega^2, \quad v = \omega R \quad\Longrightarrow\quad I = \frac{2Mgh}{\omega^2} - MR^2. \tag{1}$$

From rest with constant α, the angle turned, the distance and the angular velocity after the time t are

$$\Delta\varphi = \tfrac{1}{2}\alpha t^2, \qquad s = R\,\Delta\varphi = \tfrac{1}{2}R\alpha t^2, \qquad \omega = \alpha t. \tag{2}$$

Part 2, Newton's second law. Along the incline, and for the rotation about the centre of mass:

$$Mg\sin\theta - f = Ma, \quad fR = I\alpha, \quad a = \alpha R \quad\Longrightarrow\quad f = M(g\sin\theta - \alpha R), \quad I = \frac{MgR\sin\theta}{\alpha} - MR^2. \tag{3}$$

The iOLab. Each tape roll is a thick hollow cylinder, and the iOLab is what is left:

$$I_{\text{rolls}} = 2\cdot\tfrac{1}{2}m_{\text{roll}}\left(r^2 + R^2\right) = \tfrac{1}{2}m_{\text{rolls}}\left(r^2 + R^2\right), \qquad I_{\text{iOLab}} = I - I_{\text{rolls}}. \tag{4}$$

Expectation. Seen along y, the iOLab is a box of about $w \times t = 7.5\,\text{cm} \times 3.0\,\text{cm}$: $\frac{1}{12}m(w^2+t^2) = 1.1 \times 10^{-4}\,\text{kg m}^2$ with its mass spread evenly and $\frac{1}{12}m(w+t)^2 = 1.9 \times 10^{-4}\,\text{kg m}^2$ with all mass in its walls. Even with all its mass on the four long edges, it cannot pass $\frac{1}{4}m(w^2+t^2) = 3.3 \times 10^{-4}\,\text{kg m}^2$.

3 Method

We released the body from rest with its contact line $s = 68.2\,\text{cm}$ up the board from the low end, five times (takes 1 to 5). It rolled down, left the board at the low end and was stopped on the table. The iOLab recorded ω_y with the gyroscope at 190 Hz; the accelerometer and the magnetometer were also on but are not used. We saved a screen capture of each take and read the ω_y curve off the graph pixel by pixel in Python (one pixel is 0.011 s and 0.10 rad/s; the grid lines fix both axis scales to 0.2 %). A straight line fitted to ω against t between 2 and 18 rad/s gives α for each take; the window skips the release and the edge of the screen.

The graph shows ω only up to 20 rad/s, which the body passed after about 49 cm, so ω at the end of the ramp is not on the screen. Because $\omega(t)$ is a straight line from rest, Part 1 takes it from (2) at the moment the body leaves the board. As a check that does not use α, we also test energy conservation along the visible part of the ramp: with $\varphi = \int \omega\,dt$, (1) with $h = R\varphi\sin\theta$ makes ω^2 grow in proportion to φ, with slope $k = 2MgR\sin\theta/(I+MR^2)$.

Uncertainties: the standard error of fitted slopes, the standard deviation of the mean (SDOM) over the five takes, propagation with partial derivatives, and the standard score z. We did not calibrate the gyroscope and allow 1 % for its scale.

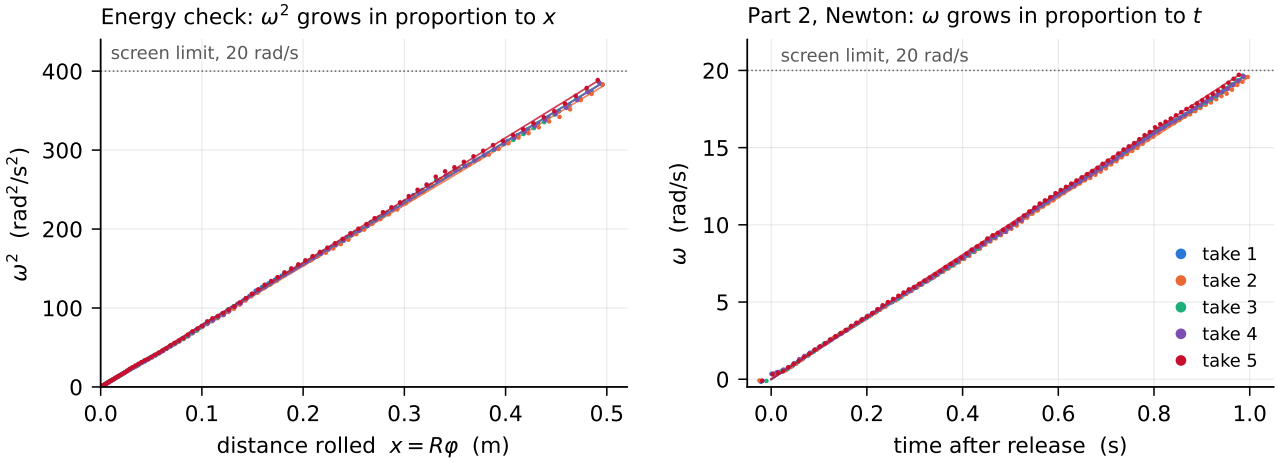


Figure 2. All five takes, read off the screen captures. Left, energy check: ω^2 against the distance rolled, $x = R\varphi$. Right, Part 2: ω against the time after the release. Lines: the fits. Dotted: the top of the app graph, 20 rad/s.

Table 2. All takes. α: slope of $\omega(t)$, fit standard error 0.03 rad/s². k: slope of ω^2 against φ, fit standard error 0.06 to 0.09 rad/s². Last column: the time after the release when the curve comes back on the screen. Last row: mean ± SDOM.

	Part 2, Newton		Energy check		
Take	α (rad/s²)	I_{iOLab} (10^{-4} kg m²)	k (rad/s²)	I_{iOLab} (10^{-4} kg m²)	back on screen (s)
1	19.85	6.58	39.65	6.61	1.24
2	19.61	6.92	39.22	6.92	1.18
3	19.87	6.55	39.64	6.62	1.18
4	19.82	6.62	39.68	6.59	1.17
5	20.16	6.15	40.31	6.15	1.16
mean	19.86 ± 0.09	6.6 ± 0.6	39.70 ± 0.17	6.6 ± 0.6	

4 Part 1: conservation of energy

The body leaves the board at the earliest moment a curve comes back on the screen, $t = (1.16 \pm 0.01)\,\text{s}$ after the release (take 5; the other takes come back 0.01 to 0.08 s later, after landing). With the mean α, (2) gives $s = \frac{1}{2}R\alpha t^2 = (68.1 \pm 1.4)\,\text{cm}$, the same as the $(68.2 \pm 0.2)\,\text{cm}$ we measured, so the body rolled without slipping. At that moment $\omega = \alpha t = (23.0 \pm 0.2)\,\text{rad/s}$, above the top of the app graph. With $h = s\sin\theta = (13.7 \pm 0.2)\,\text{cm}$, (1) gives

$$I = \frac{2Mgh}{\omega^2} - MR^2 = (1.37 \pm 0.06) \times 10^{-3}\,\text{kg m}^2, \qquad I_{\text{iOLab}} = I - I_{\text{rolls}} = (6.5 \pm 0.6) \times 10^{-4}\,\text{kg m}^2.$$

Since t and ω both come from α, putting (2) into (1) gives exactly (3): Part 1 carries the same uncertainty as Part 2 (Table 3). The energy check along the visible ramp does not use α. Every take gives a straight line through

the origin in Figure 2 (left), as energy conservation predicts, and the mean slope $k = (39.70 \pm 0.17)\,\mathrm{rad/s^2}$ gives $I_{\mathrm{iOLab}} = (6.6 \pm 0.6) \times 10^{-4}\,\mathrm{kg\,m^2}$.

5 Part 2: Newton's second law

Every take gives a straight line in Figure 2 (right): α is constant, as (3) predicts for constant forces. The mean slope $\alpha = (19.86 \pm 0.09)\,\mathrm{rad/s^2}$ gives with (3)

$$f = (0.53 \pm 0.02)\,\mathrm{N}, \qquad I = (1.37 \pm 0.06) \times 10^{-3}\,\mathrm{kg\,m^2}, \qquad I_{\mathrm{iOLab}} = I - I_{\mathrm{rolls}} = (6.6 \pm 0.6) \times 10^{-4}\,\mathrm{kg\,m^2}.$$

The normal force is $Mg\cos\theta = 5.33\,\mathrm{N}$, so rolling without slipping needs a static friction coefficient of only $f/(Mg\cos\theta) = 0.10$, well below what tape on a board gives.

6 Discussion and conclusion

The two parts agree: 6.53×10^{-4} from energy and $6.56 \times 10^{-4}\,\mathrm{kg\,m^2}$ from Newton's second law. They differ by 0.5 %, only from rounding t and ω, far below the uncertainty of $0.6 \times 10^{-4}\,\mathrm{kg\,m^2}$. The energy check along the ramp gives $6.58 \times 10^{-4}\,\mathrm{kg\,m^2}$. All three use the same recordings, so their agreement tests the analysis, not the model.

Table 3. Uncertainty budget for I_{iOLab}, the same for both parts: $|\partial I_{\mathrm{iOLab}}/\partial x|\,\delta x$, added in quadrature.

Input	angle	gyroscope scale	roll model	scatter of takes	R	r	screen reading	total
$\delta I_{\mathrm{iOLab}}$ ($10^{-5}\,\mathrm{kg\,m^2}$)	4.8	2.8	2.7	1.2	1.0	0.7	0.5	6.4

The iOLab is 23 % of $I + MR^2$ (MR^2 is 51 % and the rolls 25 %), so every 1 % in $I + MR^2$ is about 4 % in I_{iOLab}. The angle leads the budget (Table 3), then the gyroscope scale and the roll model. The roll model term comes from the cardboard core, which is lighter than tape: (4) then gives I_{rolls} about 4 % too small and I_{iOLab} up to $0.3 \times 10^{-4}\,\mathrm{kg\,m^2}$ too large.

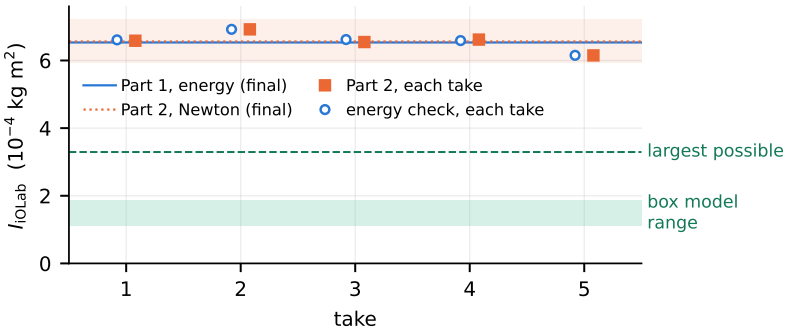


Figure 3. I_{iOLab} from every take. Lines: the final values of Part 1 and Part 2, band: their uncertainty. Green: the box model range and the largest value a 202 g body of this size can have.

Agreement does not make the result right. $I_{\mathrm{iOLab}} = 6.5 \times 10^{-4}$ to $6.6 \times 10^{-4}\,\mathrm{kg\,m^2}$ is 3.5 to 6 times the box model value and larger than $3.3 \times 10^{-4}\,\mathrm{kg\,m^2}$, the most a 202 g body of this size can have (Figure 3). A body with our masses and an iOLab inside the box range would roll down with $\alpha \approx 24.2\,\mathrm{rad/s^2}$; we measured 19.86. Something brakes the roll with about 18 % of the gravity torque. The best candidate is rolling resistance: the soft tape flattens a little where it touches the board, so the normal force acts slightly in front of the contact line. That shifted force has a torque about the centre of mass that brakes the rotation, and it takes energy out of the motion. The static friction force drops out of both parts, but rolling resistance does not, and it enters both parts in the same way. The rolling resistance coefficient this needs, resistance force over normal force, is $0.18\tan\theta = 0.04$, a normal value for a soft roll on a hard board. An error in the angle would need $\theta = 9.5°$ instead of 11.6°, far outside the 0.2° of the phone app.

Three changes would remove these errors. Runs pushed up the board and caught on the way back: going up, rolling resistance acts with gravity, going down against it, so the mean of the two slopes has no rolling resistance in it. Exported data instead of screen captures, so Part 1 can read ω at the end of the ramp directly. A calibration roll of three full turns along the table for the gyroscope scale.

Conclusion: $I_{\mathrm{iOLab}} = (6.5 \pm 0.6) \times 10^{-4}\,\mathrm{kg\,m^2}$ from conservation of energy and $(6.6 \pm 0.6) \times 10^{-4}\,\mathrm{kg\,m^2}$ from Newton's second law. Because the simple rolling model leaves out rolling resistance, this is an upper limit; for the windmill model we expect the iOLab to add 1.1 to $1.9 \times 10^{-4}\,\mathrm{kg\,m^2}$ to the wings when its y axis lies on the axis of the wings, and $I_{\mathrm{iOLab}} + m_{\mathrm{iOLab}}d^2$ when it sits a distance d from that axis.

Reflections

1. How did you approach today's experiment: did you have expectations of the final result, or did you only get the result after you had the data?
We planned the two methods before the lab and expected a value between 1.1×10^{-4} and $1.9 \times 10^{-4}\,\mathrm{kg\,m^2}$ from a box model of the iOLab. The final number came only from the data. When it came out between 3.5 and 6 times too large, we checked the angle, the radius and the ramp length before we trusted the analysis, and the comparison with the box model told us that something in the model was missing.

2. What did you learn from today's experiment?
The static friction force drops out of both methods, but rolling resistance does not, and soft tape rolls have a lot of it. Two methods that agree can both be wrong when they rest on the same assumption. The iOLab is a small part of what we measure, so small errors in the large parts grow in the answer. We also learned to export the raw data instead of saving screen captures, because the graph hid the speed at the end of the ramp.

3. What do you think you were supposed to learn from today's experiment?
To apply conservation of energy and Newton's second law for rotation to a rolling rigid body, with the parallel axis theorem, and to find the same quantity in two ways. Also to carry uncertainties through a calculation where the result is a small difference of large numbers, and to judge whether the result is physically possible.